



Automotive

We begin this track by looking at the future of the connected car; following which our speakers will discuss a realistic approach for IoT implementation in cars as well as infotronics. As the icing on the cake, we look at embedded software techniques to drive down power consumption in cars.



Medical

TeleHealth is the way to go, with doctors now being able to operate on a patient in another continent. In this track, we take a look at how mobility solutions and informatics combine to support TeleHealth at Philips, as well as how exactly the IoT enables your health and wellness. This is followed by talks on how to easily build medical electronics devices using computer on modules and a look at MedTech innovation in India.



Open source hardware

Get inspired as our speakers explain how they built exciting products like a UAV or an autopilot drone from scrap material. For those keen on making things, we have a talk to get started with your favourite -- Raspberry Pi. Not a Pi fan? Check out talks on the most exciting applications that have been built on dev boards, or on implementing Bluetooth 4.0 and sensors into your open source project.



The Great Indian Jugaad

This track celebrates clever Indian innovations (*jugaad*) in electronics engineering and design. IoT switches that control your home, a sociable robot, and a lot of other exciting innovations can be explored here!

2 > Workshops



Design your own PCBs

We have two workshops on PCB designing at eRocks. The first is for newbies. It teaches the basics about PCB design technology, and then goes 100 per cent hands-on on how to design. The second goes deeper into designing for manufacturability and covers many optimisation techniques. Both workshops will be conducted using a free PCB design tool called DesignSpark, which every attendee can get for free from the September 2014 issue of *Electronics For You* magazine.



3D printing: Your design enters the real world

A machine reminiscent of the Star Trek Replicator; a 3D printer is something that's almost magical -- creating objects out of thin air (and plastic)! This workshop gets you started with 3D printers.



Get started with the Internet of Things

The course covers the requirements for 'things' to be connected over the Internet. There will be a basic introduction to the access and networking technologies that enable IoT networks. In addition, a few fundamental design principles that need to be kept in mind when building the things and IoT networks will be discussed.



Embedded software: Get hands-on with programming ARM Cortex-M3 based devices

Microcontrollers rule! Look around and you'll find more devices getting smart every day. Ever wondered why you need to incorporate a 32-bit processing core in your embedded design, or how to go about it? We'll answer your questions about the latest Cortex-M profile of the ARM architecture and go fully hands-on with programming a Cortex M3-based device. The toolchain and platform of choice would be GNU/GCC and the STM32/TIVA series of MCUs.